

# Phase III Trial of SYN-408: Dual MET/EGFR Kinase Inhibition in Recurrent Glioblastoma

A MULTI-CENTER, RANDOMIZED, DOUBLE-BLIND CONTROLLED CLINICAL STUDY (NEXA-GBM-03)

## 1. Introduction & Clinical Rationale

**Unmet Need in Recurrent Glioblastoma (rGBM):** Glioblastoma remains the most aggressive primary brain tumor in adults. Standard first-line therapy (resection, radiotherapy, and temozolomide) yields a median overall survival (OS) of 14.6 months. Upon recurrence, effective treatment options are severely limited, with historical median OS under 9 months.

**Target Pathway: Dual MET & EGFR Co-Activation**

- EGFR Amplification / vIII Mutation:** Present in ~50% of GBM tumors, driving hyperactive downstream MAPK and PI3K signaling.
- MET Tyrosine Kinase Activation:** Serves as a primary bypass resistance mechanism under EGFR targeted monotherapy.
- SYN-408 Mechanism:** A novel, orally bioavailable small-molecule dual inhibitor designed to concurrently block active ATP-binding domains of MET and EGFR.



## 2. Trial Objectives & Hypotheses

**Primary Objective:** To evaluate whether adding SYN-408 to standard salvage chemotherapy (lomustine) improves Overall Survival (OS) compared to placebo + lomustine in patients with first-recurrent glioblastoma.

**Secondary Objectives:**

- Assess Progression-Free Survival (PFS) according to RANO 2.0 criteria.
- Determine Objective Response Rate (ORR) and Disease Control Rate (DCR).
- Characterize toxicity and safety profiles using NCI-CTCAE v5.0.
- Analyze biomarker subsets (MET high vs low expression; EGFRvIII mutant vs wild-type).

## 3. Patient Eligibility & Criteria

| Key Inclusion Criteria                                                                                                                                                                                                                                                                                                                                                                                                 | Key Exclusion Criteria                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>Histologically confirmed IDH wild-type GBM at initial diagnosis.</li><li>First radiographically confirmed recurrence (RANO criteria).</li><li>Age <math>\geq 18</math> and <math>\leq 75</math> years; KPS score <math>\geq 70</math>.</li><li>Adequate hematologic, renal, and hepatic organ reserve.</li><li>Archival tissue available for MET/EGFR biomarker assay.</li></ul> | <ul style="list-style-type: none"><li>Prior systemic therapy with specific MET or EGFR targeted kinase inhibitors.</li><li>Active brainstem metastasis or leptomeningeal disease.</li><li>Severe cardiac dysfunction (LVEF <math>&lt; 50\%</math> or QTc <math>&gt; 470</math> ms).</li><li>History of major surgical intervention within 28 days prior to Day 1.</li><li>Concurrent investigational drug administration.</li></ul> |

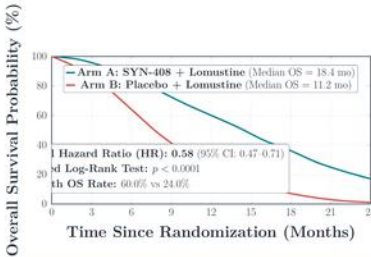
## 4. Study Design & Trial Schema



## 5. Patient Baseline Characteristics

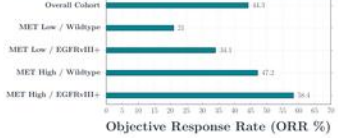
| Characteristic                              | Arm A: SYN-408 (N=210) | Arm B: Placebo (N=210) | p    |
|---------------------------------------------|------------------------|------------------------|------|
| Median Age, years (IQR)                     | 58.4 (39-66)           | 59.1 (39-67)           | 0.48 |
| Male / Female, %                            | 57.1 / 42.9            | 58.6 / 41.4            | 0.76 |
| Karnofsky Performance Score $\geq 80$ , %   | 68.6                   | 67.1                   | 0.74 |
| MGMT Promoter Methylated, %                 | 38.1                   | 37.6                   | 0.92 |
| EGFRvIII Mutation Positive, %               | 47.6                   | 46.2                   | 0.77 |
| MET Overexpression High ( $\geq 2+$ IHC), % | 52.4                   | 53.8                   | 0.77 |
| Median Time from Resection, months          | 11.8                   | 11.4                   | 0.52 |
| Prior Bevacizumab Exposure, %               | 14.3                   | 15.2                   | 0.79 |

## 6. Primary Efficacy: Kaplan-Meier Overall Survival



## 7. Biomarker Subgroup Analysis

**Enhanced Response in MET-High / EGFRvIII Biomarker Co-Positive Cohorts:**



## 8. Safety Profile & Adverse Events

| Adverse Event (CTCAE v5.0) | Arm A: SYN-408 (N=210) | Arm B: Placebo (N=210) |
|----------------------------|------------------------|------------------------|
|                            | Any Grade / Gr 3-4     | Any Grade / Gr 3-4     |
| Fatigue / Asthenia         | 34.2 / 4.3             | 48.1 / 3.8             |
| Diarrhea                   | 42.8 / 3.8             | 18.5 / 0.5             |
| Acneiform Skin Rash        | 38.6 / 2.9             | 8.1 / 0.0              |
| Nausea / Vomiting          | 33.1 / 1.9             | 31.4 / 1.4             |
| ALT / AST Elevation        | 26.2 / 5.2             | 12.4 / 1.0             |
| Thrombocytopenia           | 21.0 / 6.7             | 19.5 / 6.2             |
| Neutropenia                | 18.6 / 5.7             | 17.1 / 5.2             |
| AE Discontinuation Rate    | 4.8%                   | 3.3%                   |

## 9. Ethical Considerations & Regulatory Oversight

- Institutional Review Board (IRB) Approval:** Protocol approved by Ethics Committees of Vertex Medical Institute (Ref: VMI-IRB-2024-SSA) and Cambridge REC-B-902.
- Informed Consent:** Written informed consent was obtained from all patients or legal representatives prior to screening.
- Data Safety Monitoring Board (DSMB):** Independent semi-annual safety reviews conducted. No unexpected safety signals or trial suspension criteria were met.
- Declaration of Helsinki:** Conducted in strict accordance with ICH Good Clinical Practice (ICH-GCP) guidelines.

## 10. Conclusions & Clinical Impact

### Key Takeaways

- Significant OS Benefit:** Adding SYN-408 to lomustine reduced the risk of death by 42% (HR = 0.58,  $p < 0.0001$ ), extending median overall survival by 7.2 months.
- Robust Tumor Response:** ORR was three-fold higher in the SYN-408 arm (44.3% vs 14.8%), with maximum efficacy observed in MET-high/EGFRvIII+ co-positive patients.
- Manageable Safety Profile:** Adverse events were consistent with known EGFR/MET class effects and predominantly Grade 1-2. Discontinuation due to toxicity remained low (4.8%).
- Practice-Changing Potential:** SYN-408 represents a potential novel standard of care for biomarker-selected recurrent glioblastoma.

## 11. References & Citations